Assignment 3

Due date: Monday, November 2, 2026.
Total: 30 points
Coverage: Chapter 4: Generating Random Variables.

Question	Format	Points
1	8 multiple-choice questions on random variable generation	8
2	8 multiple-choice questions on constructing and checking random variables	8
3	6 true/false questions	6
4	2 short-answer parts with four prompts each	8
Total		**30**

Assignment PDF | Response template (.qmd)

Submission

Use **R Markdown** or **Quarto**. Submit your **source file** (`.Rmd` or `.qmd`) and **rendered PDF** to the **Assignment 3** folder on **iCollege**. Name both files `A3_LastName_FirstName` with the appropriate extension, and include your name in the PDF.
Use the response template, or follow its answer table and question labels.

- **Questions 1-2:** Select one best answer, A, B, C, or D.
- **Question 3:** Write **True** or **False**.
- **Question 4:** Give a short explanation or calculation for each prompt.

Each item is worth **1 point**. Only Question 4 requires explanations or calculations.
R code and plots are optional. If you use R, include the code in your source file. Before generating random values, set a seed (e.g., `set.seed(8670)`) and keep that line in your code. Check that your source renders to PDF without errors.

Note: The questions use the main material in Chapter 4. Topics marked **optional**, including Box-Muller, truncated-normal simulation, Wishart construction, and the real-world applications, are not required.

Notation: $N(\mu, \sigma^2)$ uses variance as its second parameter. Gamma(a, b) uses shape a and rate b.

Question 1: Generating Random Variables [8]

Select the **best answer**. Each item is worth **1 point**.

1a)

Consider the following R code:

```r
x <- numeric(100)
for (i in seq_len(100)) {
  set.seed(8670)
  x[i] <- runif(1)
}
```

What is the main problem with using `x` as 100 independent uniform draws?
A. The values must all be zero.
B. The values are sampled without replacement.
C. Resetting the seed repeats the same first draw in every iteration.
D. A uniform generator cannot be used inside a loop.

1b)

Let $U \sim \text{Uniform}(0, 1)$. Which expression generates an exponential random variable with **mean 5**?
A. $-\log(1-U)/5$
B. $-5\log(1-U)$
C. $\exp(-5U)$
D. $5U$

1c)

The Beta$(2, 2)$ density is $f(x) = 6x(1-x)$ for $0 < x < 1$, and zero otherwise. Its height at $x = 0.5$ is 1.5. Which statement is correct?
A. It cannot be a density because its height exceeds 1.
B. $P(X = 0.5) = 1.5$.
C. The area under the curve is 1.5.
D. It is a valid density: its total area is 1, even though its height can exceed 1.

1d)

An unnormalized target is $h(x) = x(1-x)$ on $(0, 1)$, with total area $1/6$. Which formula turns it into a normalized density?
A. $f(x) = 6h(x)$
B. $f(x) = h(x)/6$
C. $f(x) = h(x) - 1/6$
D. $f(x) = h(x)$

1e)

For the target in 1d), compare two rejection samplers with a uniform proposal on $(0, 1)$:

- normalized scale: target $f(y) = 6y(1 - y)$, envelope height 1.5;
- unnormalized scale: target $h(y) = y(1 - y)$, envelope height 0.25.

Suppose both samplers use the same proposal Y and the same draw $U \sim \text{Uniform}(0, 1)$, independent of Y. How do their acceptance decisions compare?
A. The normalized version accepts six times as often.
B. The unnormalized version cannot be used for rejection sampling.
C. Both accept exactly when $U \leq 4Y(1 - Y)$.
D. Both accept exactly when $U \leq Y(1 - Y)$.

1f)

Use the normalized sampler in 1e). For one proposal, $Y = 0.75$ and $U = 0.60$. Its acceptance ratio is $f(Y)/1.5 = 0.75$. What should the sampler do?
A. Accept and store 0.75.
B. Accept and store 0.60.
C. Reject because $U < 0.75$.
D. Accept and store the density height 1.125.

1g)

For the normalized sampler in 1e), a proposal is plotted at (Y, V) with $V = 1.5U$, where U is uniform on $(0, 1)$ and independent of Y. What does the point's height V represent?
A. The probability that the target equals Y.
B. The value of the target density $f(Y)$.
C. A second accepted draw from the target distribution.
D. An auxiliary random height between zero and the envelope, used to decide whether to accept Y.

1h)

An envelope lies above the target at 100 grid points. What else must be checked before using it for rejection sampling? Assume the proposal covers the target support.
A. The number of proposals is at least 100.
B. The envelope is at least as high as the target throughout its support.
C. The envelope has total area exactly 1.
D. Every proposal is accepted.

Question 2: Constructing and Checking Random Variables [8]

Select the **best answer**. Each question is worth **1 point**.

2a)

Let $X_1 \sim \text{Poisson}(1.2)$ and $X_2 \sim \text{Poisson}(2.8)$ be independent. What is the distribution of $X_1 + X_2$?
A. Poisson(1.6)
B. Poisson(3.36)
C. Poisson(4)
D. Binomial$(4, 0.5)$

2b)

Suppose $G_1 \sim \text{Gamma}(2, 4)$ and $G_2 \sim \text{Gamma}(5, 4)$ are independent, with the second parameter denoting the rate. What is the distribution of $G_1/(G_1 + G_2)$?
A. Beta$(2, 5)$
B. Beta$(5, 2)$
C. Gamma$(7, 4)$
D. Unif$(0, 1)$

2c)

Let $Z_1, \ldots, Z_6$ be independent standard normals. A programmer returns

$$W = \frac{Z_1}{\sqrt{(Z_1^2 + \cdots + Z_6^2)/6}}$$

and calls the result a t_6 draw. Which statement identifies the error?
A. The sum of six squared standard normals cannot be chi-square.
B. A t generator requires a uniform numerator.
C. Dividing the sum of squares by 6 is incorrect.
D. The numerator is not independent of the chi-square denominator.

2d)

The target is a sample of independent observations from a mixture of two bivariate normal distributions, with probabilities 0.30 and 0.70. How should the component label be generated for each observation?
A. Choose independent labels for the two coordinates.
B. Choose one label for the whole vector, then draw that vector from the selected component.
C. Draw a vector from each component and return their weighted average.
D. Choose one label once and use that component for every observation in the dataset.

2e)

A discrete random variable takes values -2, 1, and 5 with probabilities 0.20, 0.50, and 0.30. Using $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, which value is returned when $u = 0.70$?
A. 1
B. -2
C. 5
D. No value is defined at a jump.

2f)

Each row of an $n \times 2$ matrix Z contains two independent standard normal draws. Let

$$R = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

If `X <- Z %*% R`, what is the covariance matrix of a row of X?
A. $\begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}$
B. $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$
C. $\begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}$
D. $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

2g)

To generate a pair of independent standard normals, a programmer draws one $U \sim \text{Unif}(0, 1)$ and returns $X_1 = \Phi^{-1}(U)$ and $X_2 = \Phi^{-1}(U)$, where Φ is the standard normal CDF. Which conclusion is correct?
A. The pair is correct because both marginal distributions are standard normal.
B. Both marginals are correct, but the coordinates are equal and hence not independent.
C. The coordinates are independent because the quantile function is deterministic.
D. Neither coordinate has a standard normal distribution.

2h)

For independent replications with the same finite-variance model, an estimator's Monte Carlo standard error is approximately proportional to $1/\sqrt{B}$. To reduce it from 0.03 to 0.01, approximately how many replications are needed relative to the current number?
A. One third as many
B. Three times as many
C. Six times as many
D. Nine times as many

Question 3: True or False [6]

Write **True** or **False** for each statement. Each statement is worth **1 point**. No explanation is required.

3a)

With the same code and computing environment, setting the same seed at the start reproduces the same simulation results.

3b)

If X has a discrete CDF F, then $F(X)$ is necessarily uniform on $(0, 1)$.

3c)

If independent gamma variables have the same rate, their sum is gamma with the sum of their shapes and that common rate.

3d)

In R, `rnorm(100, mean = 0, sd = 2)` targets a normal distribution with variance 2.

3e)

In rejection sampling, replacing a valid bound M by a larger valid bound preserves the target distribution of accepted values, although acceptance becomes less frequent.

3f)

Matching a target distribution's mean and covariance is sufficient to establish that a multivariate generator has the correct joint distribution.

Question 4: Generating and Checking Samples [8]

Each item is worth **1 point**. Label your answers **4a.1-4a.4** and **4b.1-4b.4**. Show the requested calculations and keep explanations to one or two sentences.

4a) A normal mixture [4]

The target distribution is

$$0.25\,N(-2,1) + 0.75\,N(2,1),$$

where the second parameter is the variance. A student instead generates independent $A \sim N(-2,1)$ and $B \sim N(2,1)$ and returns

$$W = 0.25A + 0.75B.$$

1. Describe a correct procedure for generating one observation from the target mixture.
2. Find the mean of the target mixture and $E(W)$. Would checking only the mean detect the error?
3. Find the variance of the target mixture and $\mathrm{Var}(W)$.
4. Both sets of simulated values are recentered and rescaled to have mean 0 and variance 1. Does this show that they have the same distribution? Name one additional distributional check.

4b) Checking an exponential generator [4]

The intended distribution is exponential with rate 1. The exact tail probability is

$$p_0 = P(X > \log 5) = 0.20.$$

A generator produces $B = 10{,}000$ independent observations. Of these, 2,400 exceed $\log 5$, so $\hat{p} = 0.24$. All logarithms are natural. Keep full precision until the final rounding.

1. Compute the estimated MCSE, $\sqrt{\hat{p}(1-\hat{p})/B}$, to four decimal places.
2. Compute $|\hat{p} - p_0|$ divided by the MCSE, to one decimal place. Is this discrepancy small enough to dismiss as ordinary simulation variation, or should the generator be checked?
3. If the same algorithm produces 40,000 independent observations and $\hat{p}$ remains close to 0.24, what approximate MCSE would you expect? Give four decimal places.
4. Name one implementation or parameterization error worth checking. Explain why increasing B would not correct it.